

Bus Topology:

Cisco Packet Tracer – Bus Topology

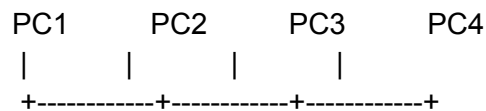
1. Open Cisco Packet Tracer

1. Open **Cisco Packet Tracer**.
2. Select the **Logical** workspace.
3. You will see a blank workspace.

2. Add Four PCs

1. At the bottom-left, click **End Devices**.
2. Select **PC**.
3. Add **4 PCs** to the workspace.
4. Arrange them in a straight line.

Example:

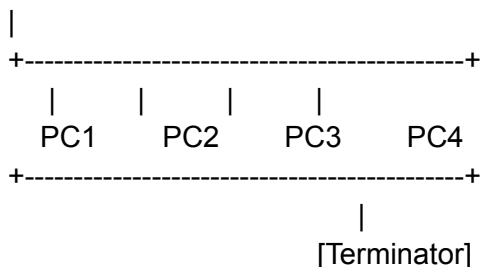


You can rename the PCs as **PC1, PC2, PC3, and PC4**.

3. Create the Bus Line

A bus topology has one main cable. All computers are connected to the same main cable.

[Terminator]



There should be a **terminator at both ends** of the bus.

4. Connect the PCs

If your Packet Tracer version provides a **Bus/Coaxial cable**:

1. Click **Connections** at the bottom-left.
2. Select the **Bus/Coaxial cable**.
3. Create the main horizontal cable.
4. Connect **PC1** to the main cable.
5. Connect **PC2** to the main cable.
6. Connect **PC3** to the main cable.
7. Connect **PC4** to the main cable.
8. Add a **terminator at both ends**.

5. Set IP Addresses

All PCs should be in the same network.

PC1

Go to:

PC1 → Desktop → IP Configuration

Enter:

IP Address: 192.168.1.1

Subnet Mask: 255.255.255.0

PC2

IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

PC3

IP Address: 192.168.1.3

Subnet Mask: 255.255.255.0

PC4

IP Address: 192.168.1.4

Subnet Mask: 255.255.255.0

For this simple practical, you do **not need to enter a Default Gateway**.

6. Check Connectivity Using Ping

Step 1 – Open PC1 Command Prompt

1. Click **PC1**.
2. Go to **Desktop**.
3. Click **Command Prompt**.

Step 2 – Ping PC2

Type:

```
ping 192.168.1.2
```

Press **Enter**.

If the connection is working, you should see:

```
Reply from 192.168.1.2  
Reply from 192.168.1.2  
Reply from 192.168.1.2  
Reply from 192.168.1.2
```

This means **PC1 can communicate with PC2**.

Step 3 – Ping PC3

From PC1, type:

```
ping 192.168.1.3
```

If you receive replies, PC1 can communicate with PC3.

Step 4 – Ping PC4

From PC1, type:

```
ping 192.168.1.4
```

If you receive replies, PC1 can communicate with PC4.

7. Test from Other PCs

From PC2

ping 192.168.1.1
ping 192.168.1.3

From PC3

ping 192.168.1.1
ping 192.168.1.4

From PC4

ping 192.168.1.1
ping 192.168.1.2

If you get **Reply from...**, the connectivity is successful.

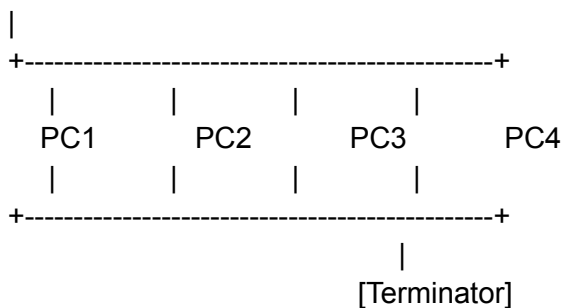
8. If Ping Fails

Check these points:

1. Check the **IP addresses**.
2. Check the **Subnet Mask**.
3. Check whether the cables are connected properly.
4. Check the **bus cable**.
5. Check the **terminators**.
6. Make sure all PCs are in the same network.

Final Bus Topology

[Terminator]



Expected Result

All PCs should be able to communicate with each other using the **ping command**.

Example:

PC1 → ping 192.168.1.2

PC1 → ping 192.168.1.3

PC1 → ping 192.168.1.4

Expected result:

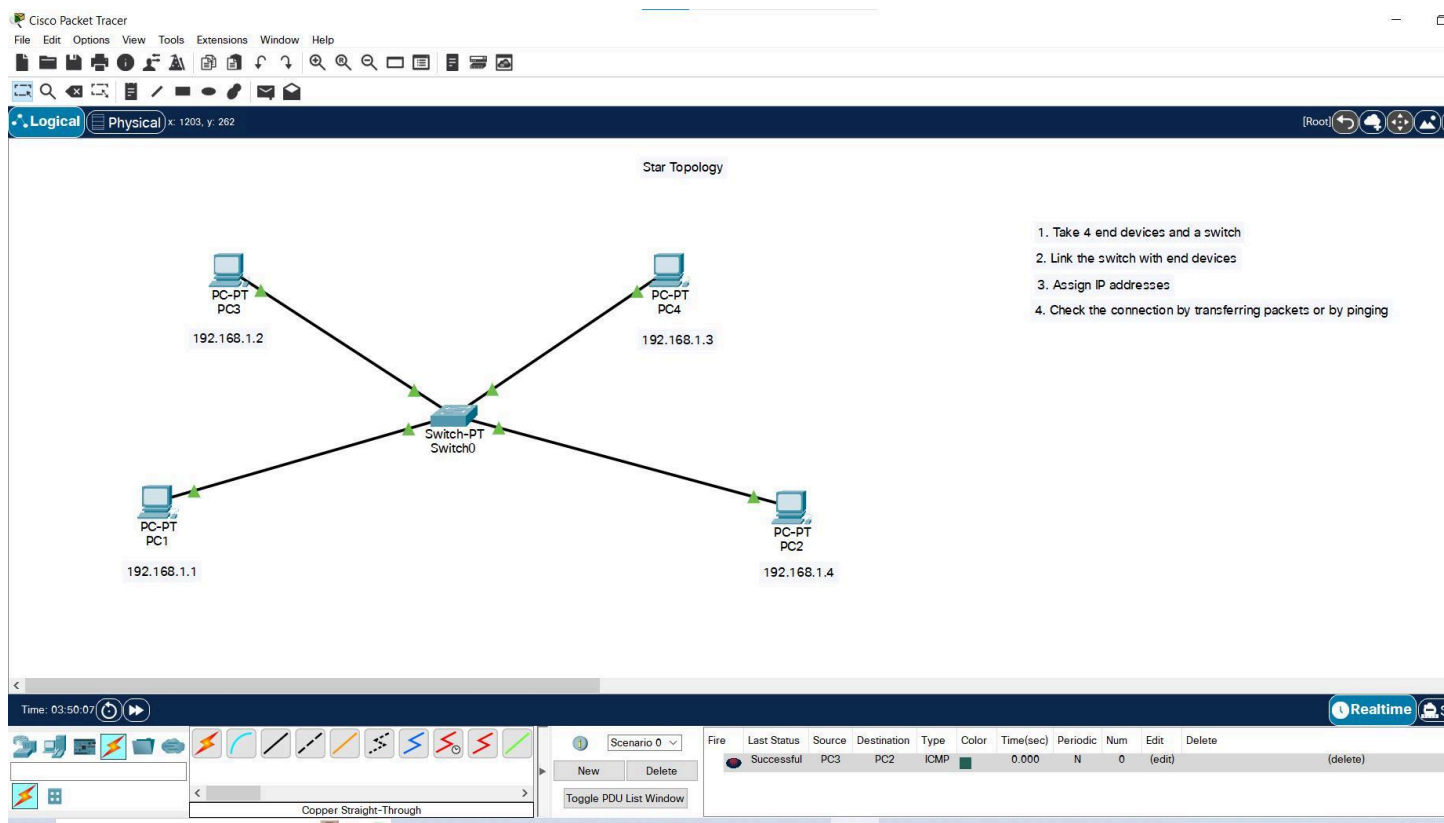
Reply from 192.168.1.x

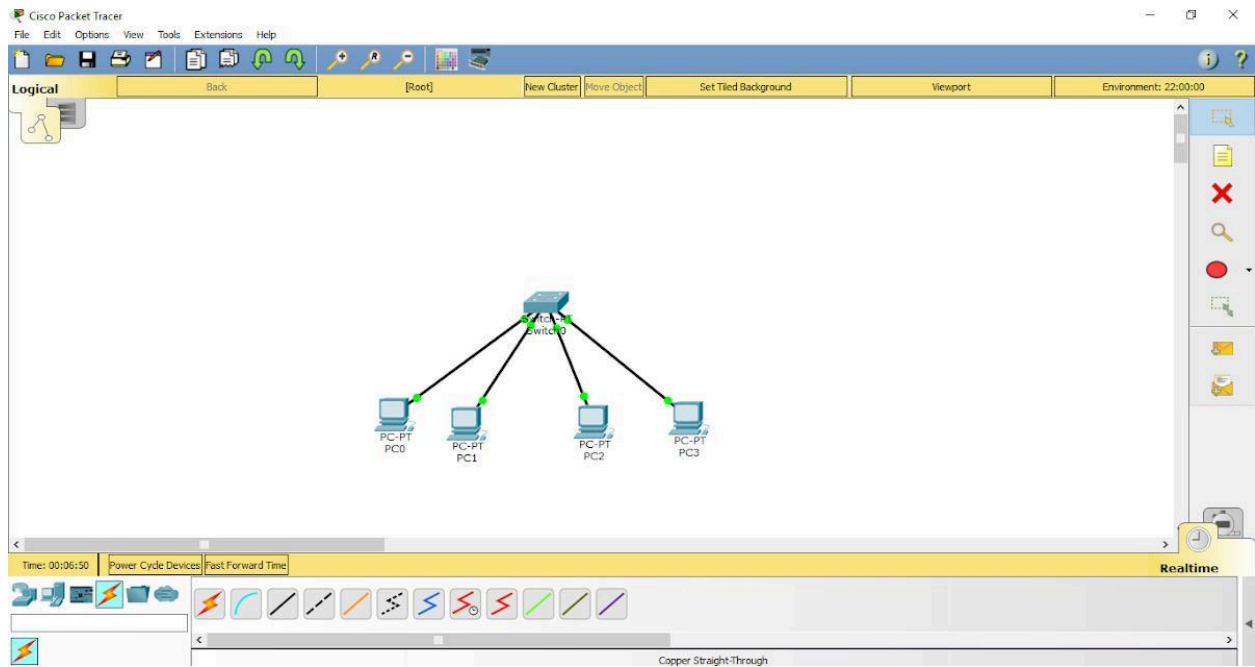
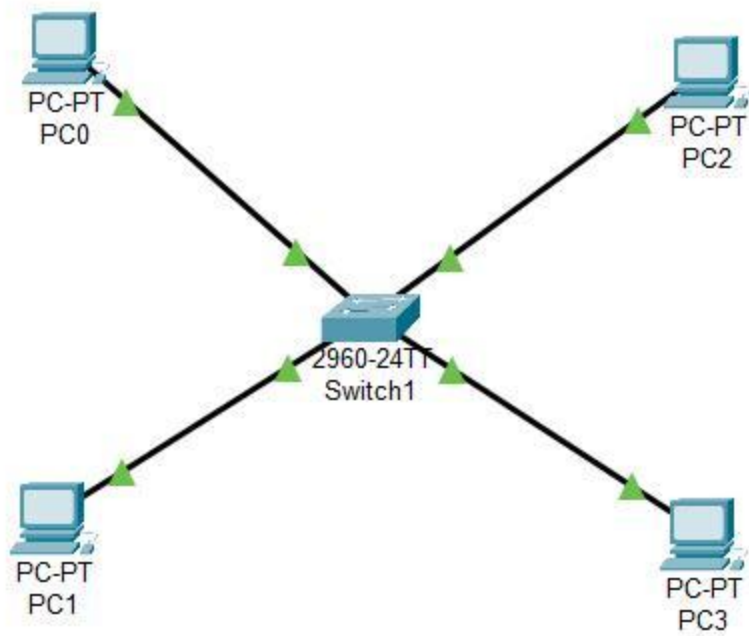
This confirms that the network connectivity is working.

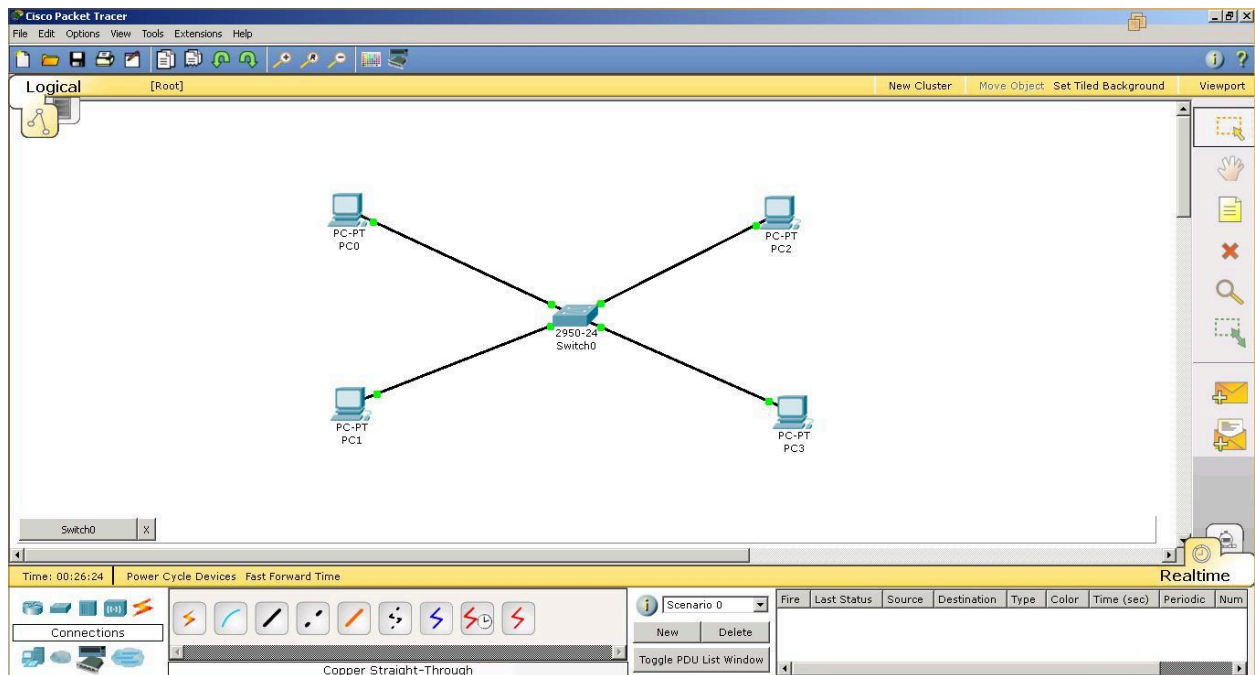
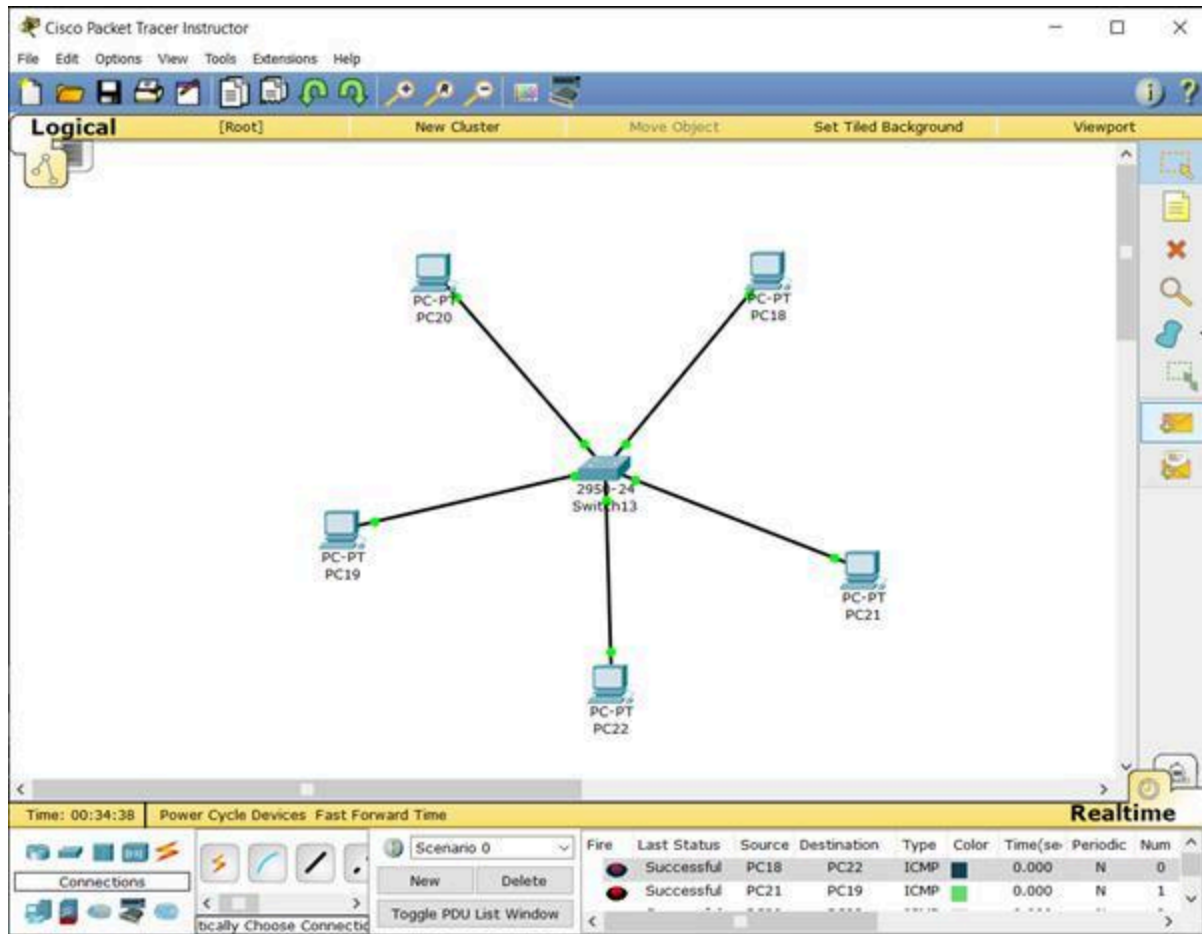
Important: In some Cisco Packet Tracer versions, a traditional **Bus/Coaxial topology** is not **directly supported**. If you send me a screenshot of your **Packet Tracer Connections** menu, I can tell you exactly which cable/icon to select in your version.

Star Topology:

Sure. Here is the **same practical for Star Topology** in simple English, with the diagram and screenshot-style guide.







Cisco Packet Tracer – Star Topology

1. Open Cisco Packet Tracer

1. Open **Cisco Packet Tracer**.
2. Select the **Logical** workspace.
3. You will see a blank workspace.

2. Add One Switch

At the bottom-left:

Network Devices → **Switches**

Select:

2960 Switch

Place it in the center of the workspace.

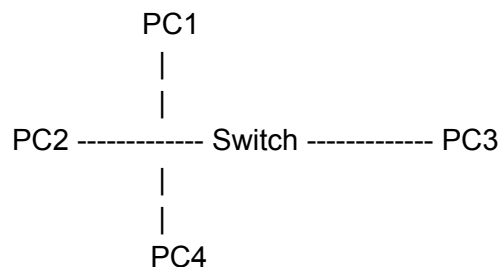
3. Add Four PCs

Go to:

End Devices → **PC**

Add **4 PCs**.

Arrange them around the switch:



This is called a **Star Topology** because all devices are connected to one central device.

4. Connect the PCs to the Switch

1. Click **Connections** at the bottom-left.
2. Select **Copper Straight-Through** cable.
3. Click **PC1**.
4. Select **FastEthernet0**.
5. Click the **Switch**.
6. Select **FastEthernet0/1**.

Now connect the other PCs:

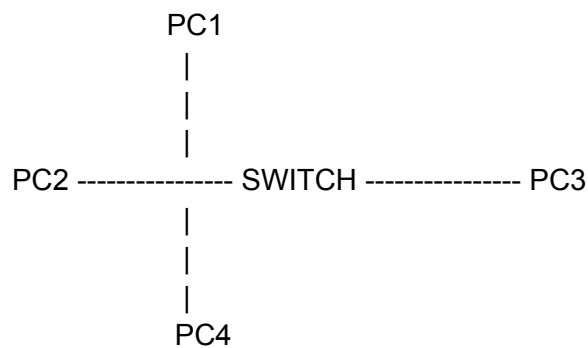
PC1 → Switch Fa0/1

PC2 → Switch Fa0/2

PC3 → Switch Fa0/3

PC4 → Switch Fa0/4

Final diagram:



5. Set IP Addresses

All PCs should be in the same network.

PC1

Go to:

PC1 → Desktop → IP Configuration

Enter:

IP Address: 192.168.1.1

Subnet Mask: 255.255.255.0

PC2

IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

PC3

IP Address: 192.168.1.3

Subnet Mask: 255.255.255.0

PC4

IP Address: 192.168.1.4

Subnet Mask: 255.255.255.0

You do not need to enter a **Default Gateway** for this simple network.

6. Check Connectivity Using Ping

From PC1 to PC2

1. Click **PC1**.
2. Go to **Desktop**.
3. Click **Command Prompt**.
4. Type:

```
ping 192.168.1.2
```

Press **Enter**.

You should get:

```
Reply from 192.168.1.2:  
Reply from 192.168.1.2:  
Reply from 192.168.1.2:  
Reply from 192.168.1.2:
```

This means **PC1** and **PC2** are connected successfully.

From PC1 to PC3

```
ping 192.168.1.3
```

From PC1 to PC4

```
ping 192.168.1.4
```

If you get **Reply from...**, the connection is working.

7. Test from Other PCs

From PC2

ping 192.168.1.1
ping 192.168.1.3

From PC3

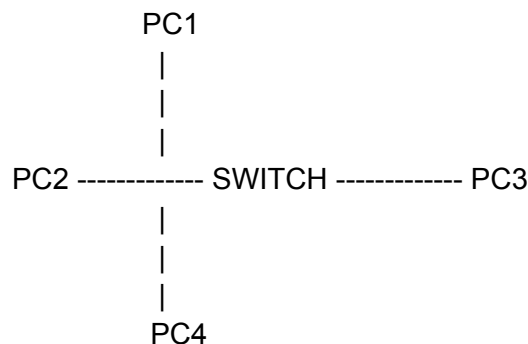
ping 192.168.1.1
ping 192.168.1.4

From PC4

ping 192.168.1.1
ping 192.168.1.2

8. Expected Result

All four PCs should be able to communicate with each other through the central switch.



If all ping commands show **Reply from...**, your **Star Topology is working successfully**.

9. If Ping Fails

Check the following:

1. Check the IP addresses.
2. Check the subnet mask.
3. Make sure you used **Copper Straight-Through** cable.
4. Check that all cables are connected to the correct switch ports.
5. Make sure the switch is connected to all four PCs.

6. Make sure all PCs are in the same network.

Port Connection Table

Device	Switch Port
--------	-------------

PC1	Fa0/1
-----	-------

PC2	Fa0/2
-----	-------

PC3	Fa0/3
-----	-------

PC4	Fa0/4
-----	-------

IP Address Table

PC	IP Address	Subnet Mask
----	------------	-------------

PC1	192.168.1.1	255.255.255.0
-----	-------------	---------------

PC2	192.168.1.2	255.255.255.0
-----	-------------	---------------

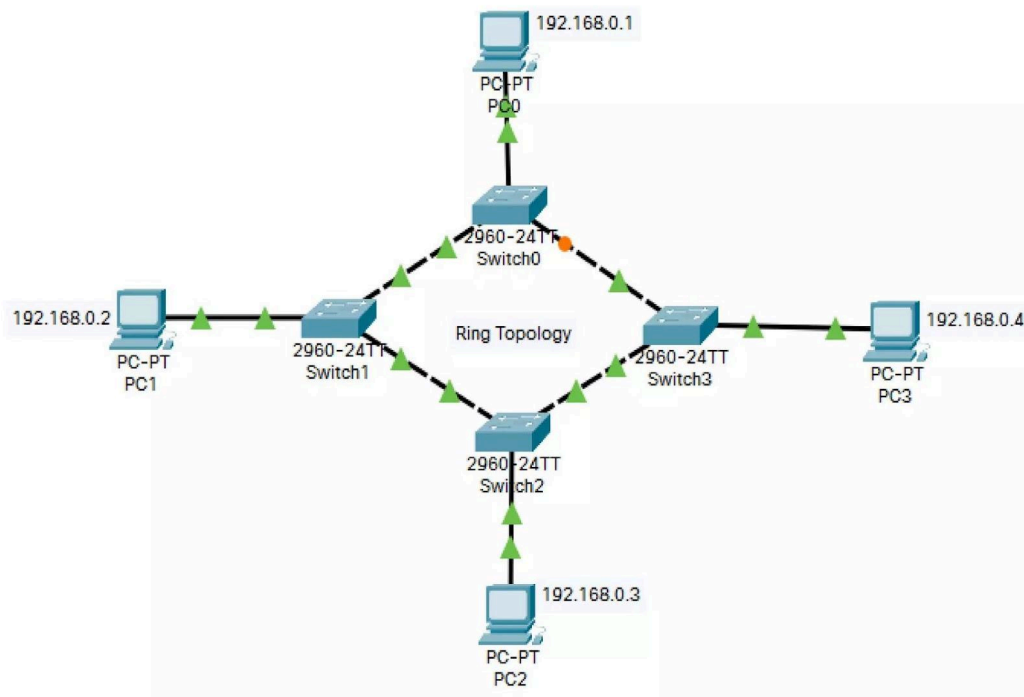
PC3	192.168.1.3	255.255.255.0
-----	-------------	---------------

PC4	192.168.1.4	255.255.255.0
-----	-------------	---------------

Screenshot guide: The generated screenshot-style diagram above shows the exact flow: **Add Switch + 4 PCs** → **Connect with Copper Straight-Through** → **Set IPs** → **Open Command Prompt** → **Ping** → **Check Reply**.

Ring Topology

Ring Topology

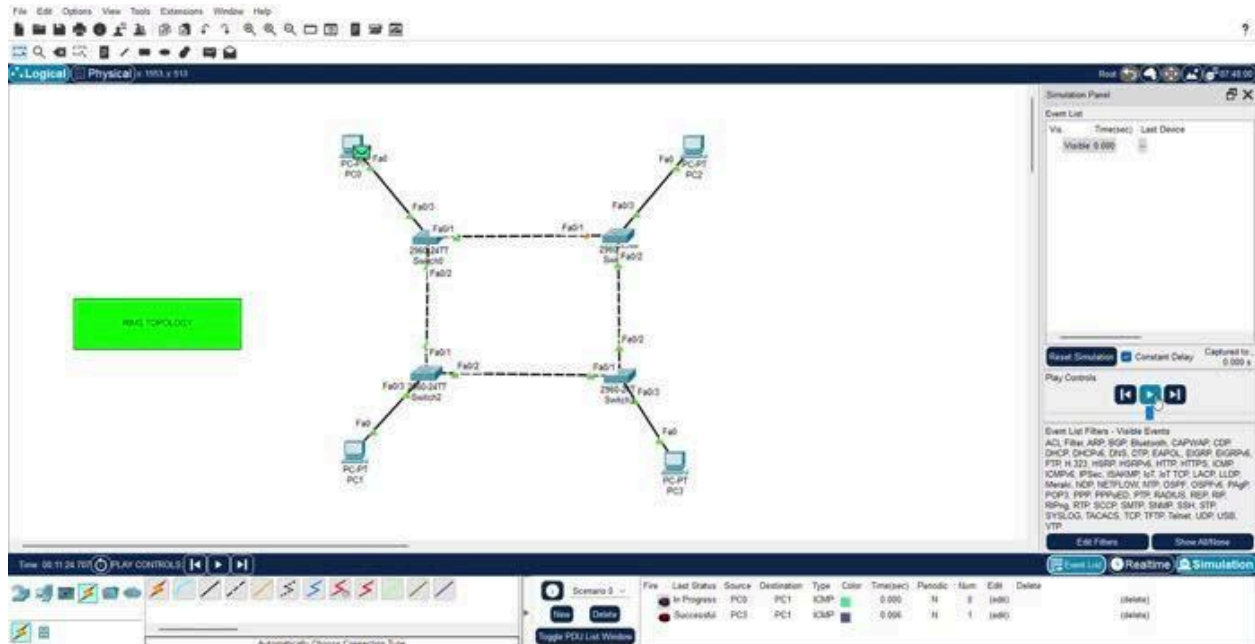


Objectives:

- Ring topology network setup with “Cisco Packet Tracer”.
- Computers are connected by a single ring-shaped cable.
- Connection between Switch to Computer.
- Connection between switch to switch.
- IP setup of computers.

IP Configurations (IP Address & Subnet mask):

- PC0 – 192.168.0.1 255.255.255.0
- PC1 – 192.168.0.2 255.255.255.0
- PC2 – 192.168.0.3 255.255.255.0
- PC3 – 192.168.0.4 255.255.255.0



Cisco Packet Tracer – Ring Topology

1. Open Cisco Packet Tracer

1. Open **Cisco Packet Tracer**.
2. Select the **Logical** workspace.
3. You will see a blank workspace.

2. Add Four Switches

Go to:

Network Devices → Switches

Select:

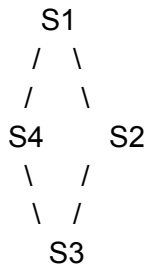
2960 Switch

Add **4 switches** to the workspace.

Rename them:

S1
S2
S3
S4

Arrange them in a square or ring shape:



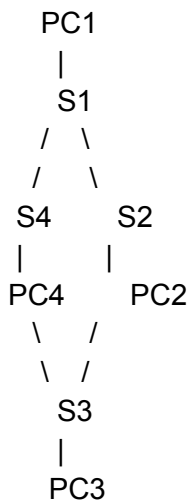
3. Add Four PCs

Go to:

End Devices → **PC**

Add **4 PCs**.

Connect one PC to each switch.



4. Connect PCs to Switches

Click:

Connections → Copper Straight-Through

Connect:

PC1 → S1

PC2 → S2

PC3 → S3

PC4 → S4

Use the following switch ports:

PC1 → S1 Fa0/1

PC2 → S2 Fa0/1

PC3 → S3 Fa0/1

PC4 → S4 Fa0/1

5. Connect the Switches in a Ring

Now connect the switches with **Copper Cross-Over** cable.

Connect:

S1 → S2

S2 → S3

S3 → S4

S4 → S1

Use these ports:

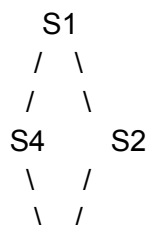
S1 Fa0/2 → S2 Fa0/2

S2 Fa0/3 → S3 Fa0/2

S3 Fa0/3 → S4 Fa0/2

S4 Fa0/3 → S1 Fa0/3

The final ring should look like:



S3

The important point is:

Every switch must be connected to two other switches.

6. Set IP Addresses

All PCs should be in the same network.

PC1

Go to:

PC1 → Desktop → IP Configuration

Enter:

IP Address: 192.168.1.1
Subnet Mask: 255.255.255.0

PC2

IP Address: 192.168.1.2
Subnet Mask: 255.255.255.0

PC3

IP Address: 192.168.1.3
Subnet Mask: 255.255.255.0

PC4

IP Address: 192.168.1.4
Subnet Mask: 255.255.255.0

For this simple network, **Default Gateway is not required.**

7. Wait for the Connections to Become Green

After connecting the devices, wait for a few seconds.

The connection indicators should become **green**.

If the links are orange for a short time, wait.

If they remain red, check the cable and port connections.

8. Check Connectivity Using Ping

Test PC1 → PC2

1. Click **PC1**.
2. Go to **Desktop**.
3. Open **Command Prompt**.
4. Type:

```
ping 192.168.1.2
```

Press **Enter**.

You should get:

```
Reply from 192.168.1.2  
Reply from 192.168.1.2  
Reply from 192.168.1.2  
Reply from 192.168.1.2
```

This means PC1 can communicate with PC2.

9. Test PC1 → PC3

From PC1 Command Prompt:

ping 192.168.1.3

You should get:

Reply from 192.168.1.3

10. Test PC1 → PC4

Type:

ping 192.168.1.4

You should get:

Reply from 192.168.1.4

11. Test from Other PCs

From PC2

ping 192.168.1.1
ping 192.168.1.3

From PC3

ping 192.168.1.1
ping 192.168.1.4

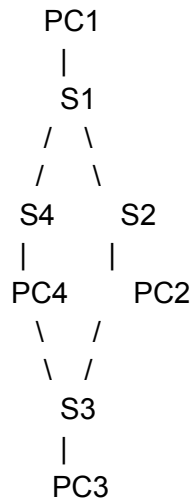
From PC4

ping 192.168.1.1
ping 192.168.1.2

If you receive **Reply from...**, the connectivity is working.

12. Final Ring Topology

Your final topology should look similar to this:



The four switches form a **closed ring**:

S1 → S2 → S3 → S4 → S1

13. Device Connection Table

Device	Connected To
PC1	S1
PC2	S2
PC3	S3
PC4	S4
S1	S2 and S4
S2	S1 and S3
S3	S2 and S4
S4	S3 and S1

14. IP Address Table

PC	IP Address	Subnet Mask
PC1	192.168.1.1	255.255.255.0
PC2	192.168.1.2	255.255.255.0
PC3	192.168.1.3	255.255.255.0
PC4	192.168.1.4	255.255.255.0

15. If Ping Fails

Check these points:

1. Check the IP addresses.
2. Check the subnet mask.
3. Check the cables.
4. Check the switch ports.
5. Make sure every switch is connected to two other switches.
6. Make sure the ring is closed.
7. Wait for the links to become green.

Expected Result

All four PCs should be able to communicate with each other.

For example:

PC1 → PC2	Successful
PC1 → PC3	Successful
PC1 → PC4	Successful
PC2 → PC3	Successful
PC2 → PC4	Successful
PC3 → PC4	Successful

A successful ping shows:

Reply from 192.168.1.x

This confirms that the **Ring Topology is working successfully.**

Mesh Topology – Cisco Packet Tracer Steps

Step 1: Add PCs

1. Open **Cisco Packet Tracer**.
2. Go to **End Devices**.
3. Select **PC**.
4. Drag and drop **4 PCs** into the workspace.
5. Rename them:
 - PC1
 - PC2
 - PC3
 - PC4

Step 2: Add Extra Ports

1. Click **PC1**.
2. Open **Physical** tab.
3. Turn **Power OFF**.
4. Add network modules to get extra Ethernet ports.
5. Turn **Power ON**.
6. Repeat the same steps for **PC2, PC3, and PC4**.

Step 3: Connect PC1 to PC2

1. Click **Connections** (lightning icon).
2. Select **Copper Cross-Over**.
3. Click **PC1** → select an available **FastEthernet port**.
4. Click **PC2** → select an available **FastEthernet port**.

Step 4: Connect PC1 to PC3

PC1 → PC3

Use another available port on both PCs.

Step 5: Connect PC1 to PC4

PC1 → PC4

Use another available port.

Step 6: Connect PC2 to PC3

PC2 → PC3

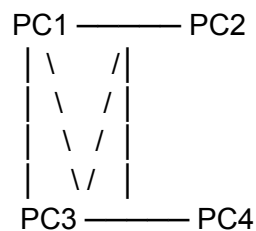
Step 7: Connect PC2 to PC4

PC2 → PC4

Step 8: Connect PC3 to PC4

PC3 → PC4

Final Connections



There should be **6 direct connections** in total.

Step 9: Test the Topology

1. Click **PC1**.
2. Go to **Desktop** → **Command Prompt**.
3. Use **ping** with the IP address of another PC.
4. Repeat the test from other PCs.
5. If you receive **Reply from...**, the connection is working.